# Chapter 13 – Course Summary

IERG4320/IEMS5726 Data Science in Practice | Cheat Sheet

## Exam Information

考试信息

**Format** (3 types):

1. **Python Programming** (30%): 3 problems – data preprocessing, ML, visualization. Fill in the blanks.
2. **Short Questions** (20%): many small problems not covered by scenario-based.
3. **Scenario-based Written** (50%): 4 problems – beginner to moderate interview questions.

**Tips**: revise assignment Q1, lecture examples, UReply questions, prepare DS interview questions, understand course project design.

## Data Science Pipeline

数据科学流程：定义问题→收集→预处理→建模→解释→进一步分析

Problem Def → Collection → Preprocessing → Model → Interpret → Further Analysis

### Data Preprocessing (8 Steps)

数据预处理8步

1. Load libraries and dataset
2. Data wrangling 数据整理
3. Check missing values 检查缺失值
4. Handle categorical values 处理分类变量
5. Handle NL, images, audio 处理文本/图像/音频
6. Dimensionality reduction 降维
7. Train/validation/test split 划分数据集
8. Feature scaling 特征缩放

### 4 Types of Data Analytics

四种数据分析

- **Descriptive**: what happened? 描述性：发生了什么
- **Diagnostic**: why did it happen? 诊断性：为什么
- **Predictive**: what will happen? 预测性：会发生什么
- **Prescriptive**: what should we do? 规范性：应该怎么做

### Basic ML Recap

机器学习基础回顾

**4 data formats**: numerical, categorical, ordinal, text/image/audio.
**4 ML problem types**: classification, regression, clustering, reinforcement learning.
**3 algorithm types**: supervised, unsupervised, reinforcement.
**3 dataset types**: training, validation, testing.

## All 17 Deep Learning Models

本课程涵盖的17个深度学习模型

1. **MLP** – basic classification/regression 多层感知机
2. **DQN** – basic RL agent 深度Q网络
3. **DeepCoNN** – recommender (review CNN) 深度协同神经网络
4. **Wide & Deep** – recommender (memorize + generalize) 宽深网络
5. **GCN** – recommender (graph convolution) 图卷积网络
6. **RNN** – sequence modeling 循环神经网络
7. **LSTM** – sequence with gates 长短期记忆
8. **BERT** – encoder Transformer (NLP) 双向编码器
9. **GPT** – decoder Transformer (text gen) 生成预训练Transformer
10. **Tacotron 2.0** – speech synthesis (TTS) 语音合成
11. **HuBERT** – speech recognition (STT) 语音识别
12. **CNN** – image captioning encoder 卷积神经网络
13. **GAN** – image generation 生成对抗网络
14. **Diffusion Model** – image generation 扩散模型
15. **D3QN** – game agent 对决双重DQN
16. **A3C** – game agent (actor-critic) 异步优势Actor-Critic
17. **AlphaZero** – game agent (MCTS + self-play) 自我对弈

Plus: **TripoSR** (2D→3D) and **Shap-E** (text→3D) for 3D generation.

## Chapter-by-Chapter Quick Reference

各章快速索引

### Ch2–5: Foundations

**Data types**: structured (tabular) vs unstructured (text/image/audio).
**Preprocessing**: missing values (drop/impute), encoding (one-hot, label), scaling (StandardScaler, MinMaxScaler), PCA for dim reduction.
**Data representation**: TF-IDF (text), embeddings (word2vec), spectrograms (audio), pixel arrays (image).

### Ch6: ML – Classification & Regression

**Classification**: DT, RF, SVM, KNN, ANN. Metrics: accuracy, precision, recall, F1, MCC, AUC-ROC.
**Regression**: Linear, Logistic, Ridge, LASSO, SVR, ANN. Metrics: $R^2$, MAE, RMSE.
**Model selection**: k-fold CV, train/val/test split.

### Ch7: Clustering & RL

**Clustering**: K-Means (elbow method), Agglomerative (linkage types), Spectral.
**Metrics**: Silhouette $[-1, 1]$ ↑, Calinski-Harabasz ↑, Davies-Bouldin ↓.
**RL**: agent/env/state/action/reward, Q-Learning, $\varepsilon$-greedy.

### Ch8: Data Visualization

**Preprocessing**: histogram, boxplot, scatter, Q-Q, heatmap, pairplot.
**Analysis**: ROC, training curve, residual, calibration, learning curve.
**Combining**: subplots, overlay, shared axes. Text: word cloud.
**7 best practices**: audience, simplicity, color, labels, scale, context, honesty.

### Ch9: Business – RecSys & Stock

**RecSys 3 generations**:
1st: CF (user/item), content-based, knowledge-based.
2nd: hybrid, matrix factorization ($R \approx PQ^T$).
3rd: DeepCoNN, Wide&Deep, GCN.
**Stock**: yfinance, EMA, LSTM (sliding window), DQN trading.
**Metrics**: Sharpe ratio, max drawdown, Sortino ratio.

### Ch10: NLP – Semantic & Text Gen

**Semantic**: TF-IDF+NB (0.84), NBoW (0.86), LSTM, BERT (MLM+NSP), Transformer (TinyBERT 0.86).
**BERT variants**: ALBERT, RoBERTa, DistilBERT, TinyBERT.
**Benchmarks**: GLUE (9 tasks), SQuAD (QA).
**Text gen**: Markov chain, GPT (1→4), LLaMA.
**Architecture**: encoder-only (BERT) vs decoder-only (GPT) vs seq2seq (T5).

### Ch11: Multimedia

**Audio**: vocal separation (librosa, Demucs).
**TTS**: SPSS, concatenative, WaveNet (dilated causal conv), Tacotron2. Metric: MOS.
**STT**: Wav2Vec 2.0, HuBERT (k-means pseudo-labels).
**Image caption**: CNN-RNN+attention, ViT+Transformer. Metrics: BLEU, METEOR, CIDEr, SPICE.
**Image gen**: GAN, DALL-E+CLIP, diffusion. Metric: FID.
**Video**: DeepFake (encoder-decoder swap).

### Ch12: Game

**Blackjack**: state=(sum, dealer, ace), hit/stand.
**D3QN**: double (reduce overestimation) + dueling ($V(s) + A(s,a)$).
**A3C**: async parallel workers, actor-critic, advantage.
**AlphaZero**: MCTS + dual-headed net + self-play.
**EWC**: prevent catastrophic forgetting across games.
**3D gen**: TripoSR (image→3D), Shap-E (text→3D).
**Robot**: transfer learning, imitation, behavior cloning, inverse RL, diffusion policies.

## Future of Data Science

数据科学的未来

**Roles**: Data Scientist (models) vs ML Engineer (deploy) vs Data Engineer (pipeline/infra).
**Emerging**: IoT (smart sensors), spatial intelligence (3D reasoning), Vision-Language Models (VLMs, multimodal).
**Final message**: embrace GenAI tools; think products over applications; specialized agents over general ones.

**Course in One Line**:
Data pipeline (collect → preprocess → model → interpret) applied to business, NLP, multimedia, and games using 17 deep learning models from MLP to diffusion.
数据流水线应用于商业/NLP/多媒体/游戏，涵盖17个深度学习模型